

Week 1: Introduction & Progress Review

Roadmap, deliverables, grading, data sources, and resources

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- ▶ Understand the arc of the program and what you will build
- ▶ Be clear on the timeline, the three milestones, and how you are graded
- ▶ Know where the data comes from (Coincall) and how to pull it
- ▶ Have the Python + PyTorch and C++/pybind11 toolchains running

- ▶ Maker vs. taker: we build the MAKER – continuously quoting both sides
- ▶ The spine of the course is the Avellaneda-Stoikov model
- ▶ Week 1 status: environment setup, data access, and this orientation
- ▶ By Week 5 you ship a calibrated quoting engine (Milestone 1)

- ▶ Problem sets: Python + PyTorch (autograd does differentiation for you)
- ▶ Fast computation (Week 7): C++ exposed to Python via pybind11
- ▶ Mentorship is asynchronous (email/Telegram); tutors run day-to-day
- ▶ Office hours with tutors; come with a minimal reproducible snippet

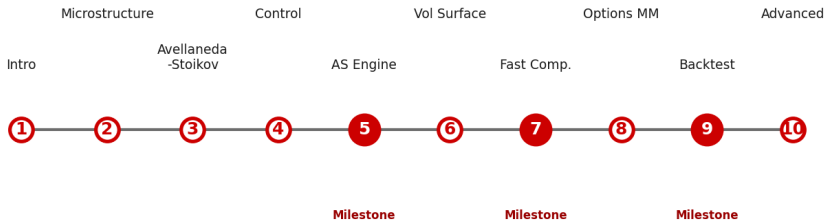
- ▶ 10 weeks. Week 1 introduction (done). Weeks 2-10 technical.
- ▶ Milestone 1 (Wk 5): Avellaneda-Stoikov engine, Python/PyTorch
- ▶ Milestone 2 (Wk 7): fast pricing/Greeks library, C++ via pybind11
- ▶ Milestone 3 (Wk 9): full options market-making backtest
- ▶ Final (Wk 10): one advanced extension + report (≤ 30 pp) + 20-min talk
- ▶ Problem sets are due Sunday 23:59 ET; milestones Friday of the week

- ▶ Problem sets 30% | Milestones (3) 30% | Final 30% | Participation 10%
- ▶ Code graded on correctness, reproducibility, quality – in that order
- ▶ Numerical-verification problems: right setup + right error magnitude
- ▶ Milestone 2 latency is a HARD GATE (sub-5ms surface) – caps at 70 if missed
- ▶ Late PS: -20% until Tue; nothing after. Milestones: no late submission.
- ▶ Set random seeds: reproducibility is explicitly graded

- ▶ Exchange: Coincall. REST base: <https://api.coincall.com>
- ▶ Provided as Parquet: BTC/ETH options L2 + trades, perp L2, funding, SVI
- ▶ Coincall orderbook is L2 (aggregated depth) – there is NO L3 feed
- ▶ You can also pull live via the REST/WebSocket API (next slides)
- ▶ Auth: API key + HMAC-SHA256 signature; market data is largely public
- ▶ Always cross-check exact endpoints at <https://docs.coincall.com/>

- ▶ Stochastic control: Pham (2009), Ch. 3-4 (HJB, verification)
- ▶ Market making theory: Gueant (2016), Ch. 4; Cartea et al., Ch. 10-11
- ▶ Options/vol surface: Gatheral (2006); Gatheral-Jacquier (2014)
- ▶ Hedging: Taleb (1997), gamma scalping & the gamma-theta identity
- ▶ El Aoud-Abergel (2015): multi-Greek option market making (Week 8)
- ▶ We DERIVE the key results in lecture and explain every modeling choice

Program Roadmap — 10 Weeks



MAKER (this course): quote both sides, earn the spread

TAKER (Wk 9 guest): cross the spread to execute

- ▶ Base URL: <https://api.coincall.com> | Python SDK: github.com/CoincallExchange
- ▶ Signed request headers: X-CC-APIKEY, sign (HMAC-SHA256, upper hex), ts (ms), X-REQ-TS-DIFF
- ▶ Signing: sort params alphabetically, append uuid, ts, x-req-ts-diff, HMAC-SHA256 with secret
- ▶ Rate limits (per docs): orderbook 30/2s per user; public config 10/2s per IP
- ▶ Key endpoints you will use across the program:
 - ▶ Instruments: GET /open/option/getInstruments/{baseCurrency}
 - ▶ Option chain: GET /open/option/get/v1/{index}
 - ▶ Option book: GET /open/option/order/orderbook/v1/{symbol} (100 depth)
 - ▶ Option detail: GET /open/option/detail/v1/{symbol} (mark, IV, delta/gamma/vega/theta)
 - ▶ Trades: GET /open/option/trade/lasttrade/v1/{symbol}
 - ▶ Futures book: GET /open/futures/order/orderbook/v1/{symbol}
 - ▶ Funding: GET /open/public/fundingRate/v1?symbol=BTCUSD
 - ▶ Klines: GET /open/option/market/kline/history/v1/{optionName}?period=h1
- ▶ WebSocket channels: /options/pricing, /options/orderbook, /options/lasttrade, /options/kline
- ▶ ALWAYS verify the exact path/params against <https://docs.coincall.com/> – the API

```

import time, hmac, hashlib, uuid, requests

BASE = "https://api.coincall.com"

def signed_get(path, params=None, api_key="", api_secret=""):
    """Minimal Coincall signed GET. Public market-data endpoints often work
    unsigned; this shows the signed pattern used for account endpoints."""
    params = params or {}
    ts, window = str(int(time.time() * 1000)), "5000"
    items = sorted(params.items()) # sort alphabetically
    payload = "&".join(f"{k}={v}" for k, v in items)
    payload += f"&uuid={uuid.uuid4().hex}&ts={ts}&x-req-ts-diff={window}"
    sign = hmac.new(api_secret.encode(), payload.encode(),
                    hashlib.sha256).hexdigest().upper()
    headers = {"X-CC-APIKEY": api_key, "sign": sign,
              "ts": ts, "X-REQ-TS-DIFF": window}
    r = requests.get(BASE + path, params=params, headers=headers, timeout=10)
    r.raise_for_status()
    return r.json()

# Example: list all BTC option instruments
print(signed_get("/open/option/getInstruments/BTC"))

```

Market maker liquidity provider that continuously quotes bid and ask, earning the spread

Taker participant that crosses the spread for immediate execution

Bid-ask spread best ask minus best bid; the maker's revenue and risk premium

Inventory the maker's net position; the main source of risk

Milestone graded software deliverable, due in Weeks 5, 7, and 9

PyTorch / autograd Python ML library; automatic differentiation computes Greeks for you

pybind11 library that exposes C++ functions to Python (used in Week 7)

- ▶ Cartea-Jaikumar-Penalva, Ch. 1; Avellaneda-Stoikov (2008) – skim
- ▶ Coincall API docs: <https://docs.coincall.com/> (read 'Getting Started')

- ▶ Avellaneda, M., & Stoikov, S. (2008). High-frequency trading in a limit order book. *Quantitative Finance*, 8(3), 217-224.
- ▶ Gueant, O. (2016). *The Financial Mathematics of Market Liquidity*. Chapman & Hall/CRC.
- ▶ Cartea, A., Jaikumar, S., & Penalva, J. (2015). *Algorithmic and High-Frequency Trading*. Cambridge Univ. Press.
- ▶ Coincall API documentation. <https://docs.coincall.com/>